

The CSV files given were large in size, and initial attempts to clean the data using Python resulted in errors because the processes exceeded the RAM limit of Colab. To resolve this, I switched the runtime to R, which handled the memory load efficiently, allowing me to merge the cleaned datasets into a single master file. Additionally, the provided GeoJSON file was retained not as tabular data, but used as a spatial visualization blueprint for mapping. I only employed standard colors on the dashboards and ensured the user could interact with the visualizations. I structured my dashboard narrative depending on what they visualize, moving logically from high-level observations to smaller market mechanics. The presentation begins with basic trends and calculations, which are seen as simple but provide immediate, useful information. The subsequent Analysis dashboards are placed in the latter part of the narrative because they dive deeper than the first part, serving as an explanation for the initial trends. Finally, metrics regarding availability and minimum nights conclude the story as supplementary, extra information.

Key Insights and Dashboard Contributions

Dashboard 1: Seasonal Pricing and Occupancy Trends

This dashboard establishes the macro-level market rhythm. Initially, the data points for average prices and occupancy had very small differences from each other. This caused the trends to look like a straight line when the axis starts from zero. I unchecked the box that makes it start from zero, which revealed the zigzag line. This adjustment successfully shown clear peaks and valleys in pricing and booking rates throughout the year.

Dashboard 2: Host Performance

This contributes to the narrative by showing whether the market is dominated by a few mega-hosts or distributed among individual property owners.

Dashboard 3 & 4: Average Price and Rentals Analysis

These dashboards represent the deeper analysis section. By mapping average prices and rental distributions across London's neighbourhoods, these visualizations explain the financial trends seen in Dashboard 1. They reveal which specific boroughs drive the highest revenue and how different room types are dispersed across the city.

Dashboard 5: Reviews Analysis

This dashboard tracks historical market growth and regional popularity. By visualizing reviews over time and a heatmap, it identifies which areas have historically sustained the most foot traffic and guest interaction.

Dashboard 6: Availability & Minimum Nights

Placed at the end of the narrative, this dashboard provides extra operational information. It categorizes listings by their rental periods and compares how these minimum night requirements impact the average price, giving a final, practical view of host strategies.

Assumptions and Limitations

Data Aggregation Assumptions: It is assumed that the review counts act as a reliable proxy for actual booking volume, as not every guest leaves a review. Furthermore, the visualizations assume that the standard color palettes chosen are universally accessible and sufficient for interpreting the data without needing complex colors.

Hardware and Formatting Limitations: The analysis was initially limited by hardware constraints in Google Colab, where I switched to R to complete the ETL pipeline. Additionally, the visualization phase was limited by Tableau's default axis scaling; because seasonal price swings in a massive city like London are thinned when averaged, manual axis manipulation was strictly required to prevent the data from hiding its own trends. Finally, there is an assumption that all boolean logic changes from Airbnb's raw data (such as formatting shifts from 'f' to 'FALSE') were fully caught and standardized during the R data preparation phase.